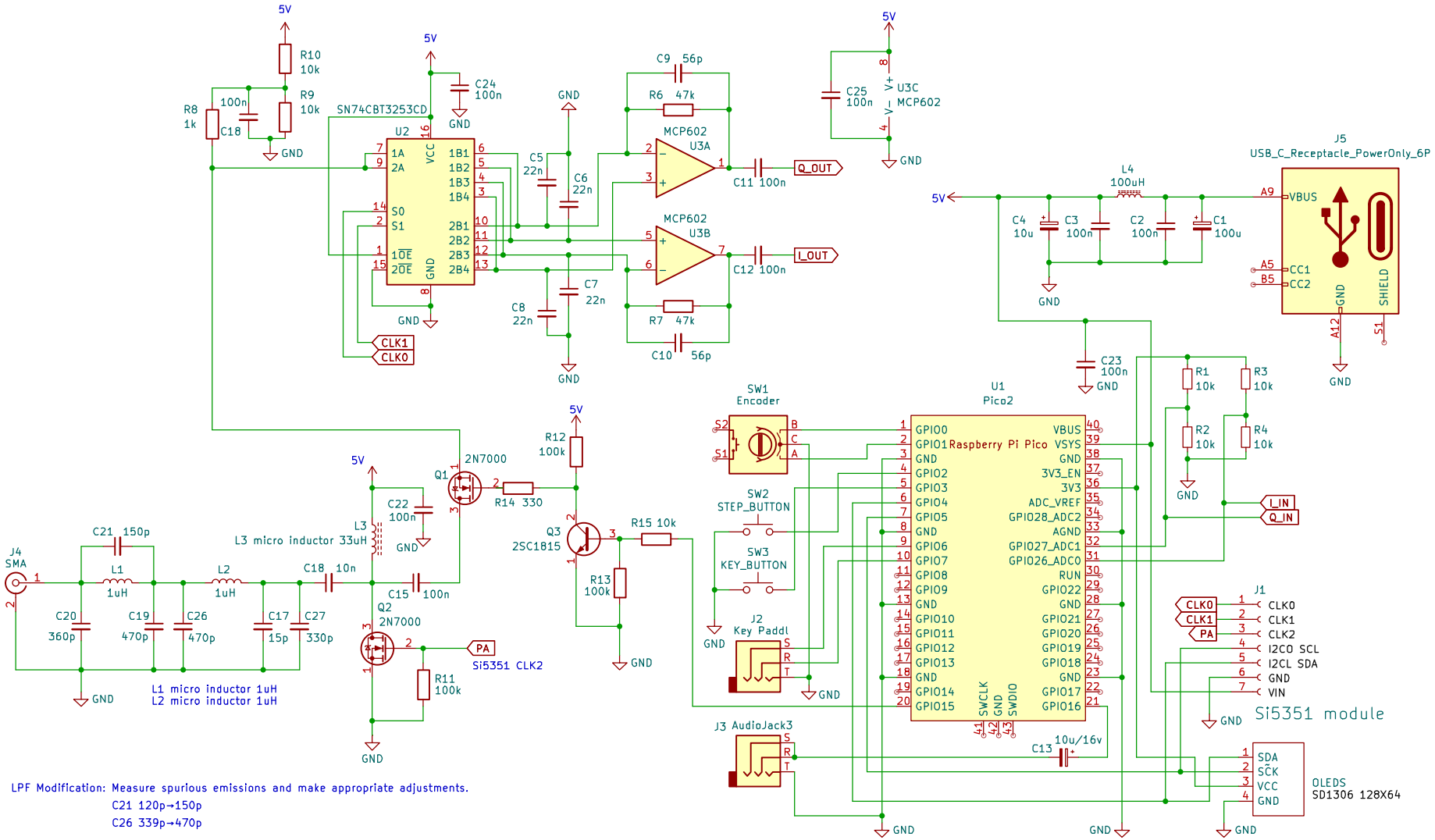
5V
R10 10k
R9 10k
R8 1k
C18 100n
GND
SN74CBT3253CD
U2
C24 100n
C9 56p
R6 47k
MCP602
U3A
C11 100n
Q_OUT
C5 22n
C6 22n
C7 22n
C8 22n
MCP602
U3B
C12 100n
I_OUT
R7 47k
C10 56p
C25 100n
U3C
MCP602
CLK1
CLK0
J5
USB_C_Receptacle_PowerOnly_6P
L4 100uH
C4 10u
C3 100n
C2 100n
C1 100u
VBUS
CC1
CC2
SHIELD
C23 100n
R1 10k
R2 10k
R3 10k
R4 10k
I_IN
Q_IN
SW1 Encoder
SW2 STEP_BUTTON
SW3 KEY_BUTTON
J2 Key Paddl
J3 AudioJack3
U1 Pico2
Raspberry Pi Pico
C13 10u/16v
J1
CLK0
CLK1
CLK2
I2C0 SCL
I2C1 SDA
GND
VIN
Si5351 module
OLEDS SD1306 128X64
SDA
SCK
VCC
2N7000
Q1
R14 330
R12 100k
Q3 2SC1815
R15 10k
R13 100k
L3 micro inductor 33uH
C22 100n
C15 100n
C21 150p
C18 10n
L1 1uH
L2 1uH
C20 360p
C19 470p
C26 470p
C17 15p
C27 330p
J4 SMA
Q2 2N7000
R11 100k
PA
Si5351 CLK2
L1 micro inductor 1uH
L2 micro inductor 1uH
LPF Modification: Measure spurious emissions and make appropriate adjustments.
C21 120p→150p
C26 339p→470p
Class E amplifier: C17 affects operation.
Replace the toroidal core with a microinductor 5V 0.43W efficiency77%
JR3XNW
Sheet: /
File: pico_CW_TRX-microL.kicad_sch
Title: pico2 CW TRX
Size: A4
Date: 2026-03-30
Rev:
KiCad E.D.A. 9.0.7
Id: 1/1